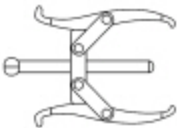
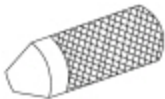
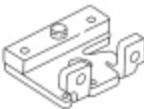
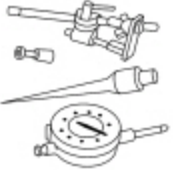
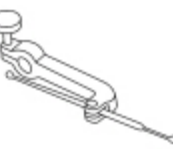
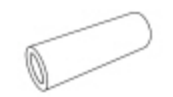


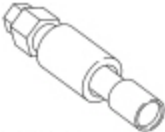
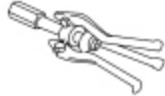
Differential Ring And Pinion



Special Tool(s)

 ST1321-A	2 Jaw Puller 205-D072 (D97L-4221-A) or equivalent
 ST1743-A	Adapter for 205-S127 205-105 (T76P-4020-A3)
 ST1429-A	Adapter for 205-S127 205-109 (T76P-4020-A9)
 ST1431-A	Adapter for 205-S127 205-110 (T76P-4020-A10)
 ST1432-A	Adapter for 205-S127 205-111 (T76P-4020-A11)
 ST1433-B	Adapter for 205-S127 205-129 (T79P-4020-A18)
 ST1150-B	Adapter for Differential Housing Spreader 205-368 (T96T-4000-A)

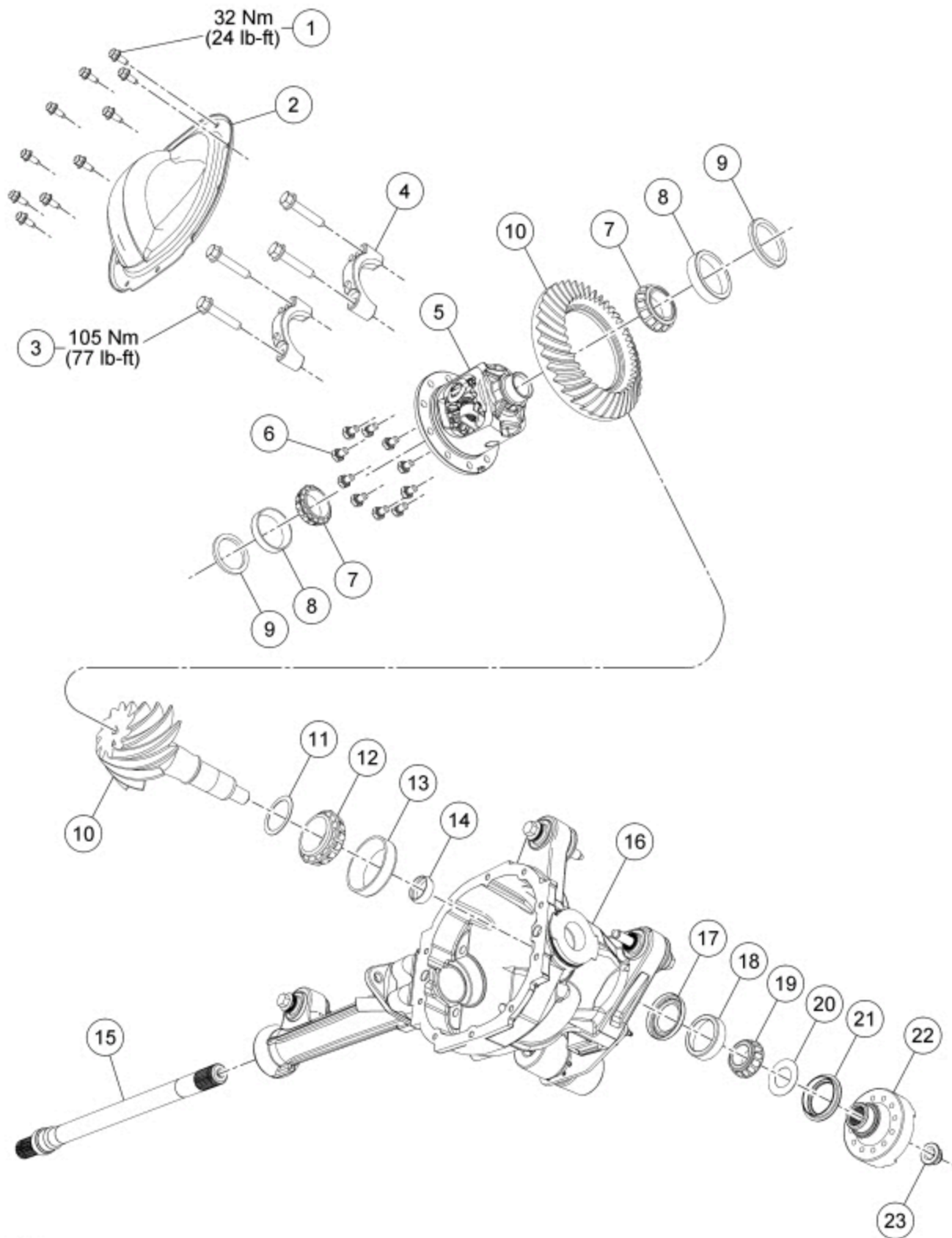
 ST1214-A	Dial Indicator Gauge with Holding Fixture 100-002 (TOOL-4201-C)
 ST1348-A	Gauge, Clutch Housing 308-021 (T75L-4201-A)
 ST1434-A	Gauge Tube, Drive Pinion 205-336 (T93P-4020-A)
 ST1257-A	Holding Fixture, Drive Pinion Flange 205-126 (T78P-4851-A)
 ST1186-A	Holding Fixture, Transmission 307-003 (T57L-500-B)
 ST1357-A	Installer, Differential Carrier Bearings 205-D044 (D81T-4221-A) or equivalent
 ST1437-A	Installer, Differential Side Bearing 205-010 (T57L-4221-A2)
 ST1484-A	Installer, Drive Pinion Bearing Cone 205-005 (T53T-4621-C)
 ST1361-A	Installer, Drive Pinion Bearing Cup 205-024 (T67P-4616-A)

 ST1800-A	Installer, Drive Pinion Flange 205-002 (TOOL-4858-E)
 ST1358-A	Installer, Drive Pinion Oil Seal 205-208 (T83P-4676-A)
 ST1882-A	Protector, Drive Pinion Thread 205-460
 ST1200-A	Remover, Bearing Cup 308-047 (T77F-1102-A)
 ST2418-A	Remover, Transmission Fluid Pump 307-397
 ST1185-A	Slide Hammer 100-001 (T50T-100-A)
 ST1259-A	Spreader, Differential Carrier 205-001 (TOOL-4000-E)
 ST1344-A	Spreader, Differential Housing (Plate) 205-335 (T93P-4000-A) (RH)
 ST2043-A	Step Plate 205-D061 (D83T-4205-C2) or equivalent

Material

Item	Specification

Motorcraft® High Contrast Hypoid Gear Marking Compound XG-14	—
Motorcraft® SAE 80W-90 Premium Rear Axle Lubricant (US); Motorcraft® SAE 80W-90 Premium Axle Lubricant (Canada) XY-80W90-QL (US); CXY-80W90-1L (Canada)	WSP- M2C197-A
Motorcraft® Premium Long-Life Grease XG-1-E1	ESA-M1C75- B
Motorcraft® Silicone Gasket and Sealant TA-30	WSE- M4G323-A4



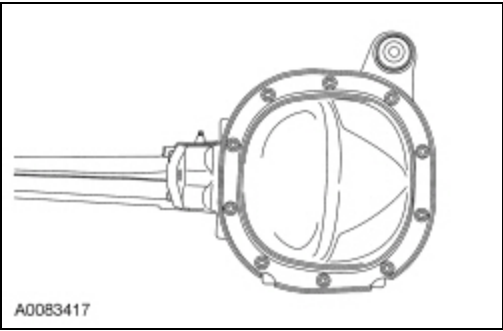
N0103507

Item	Part Number	Description
1	390919 390919	Differential housing cover bolt (10 required)
2	3N136 3N136	Differential housing cover
3	—	Differential bearing cap bolt (part of 3A412) (4 required)
4	—	Differential bearing cap (part of 3A412) (2 required)
5	3C202 3C202	Differential assembly
6	388056 388056	Ring gear bolt (10 required)
7	4221 4221	Differential bearings (2 required)

8	4221 4221	Differential bearing cups (2 required)
9	4A451 4A451	Differential bearing shims (2 required)
10	—	Ring gear (part of 4209)
10	—	Pinion gear (part of 4209)
11	4663 4663	Drive pinion bearing adjustment shim
12	4628 4628	Inner drive pinion bearing cup
13	4630 4630	Inner drive pinion bearing
14	4662 4662	Drive pinion collapsible spacer
15	3A380 3A380	Intermediate axle shaft
16	3A412 3A412	Differential housing
17	4631 4631	Drive pinion oil baffle
18	4616 4616	Drive pinion outer bearing cup
19	4621 4621	Drive pinion outer bearing
20	386989 386989	Drive pinion oil slinger
21	3N134 3N134	Drive pinion oil seal
22	4858 4858	Drive pinion flange
23	389546 389546	Drive pinion nut and washer

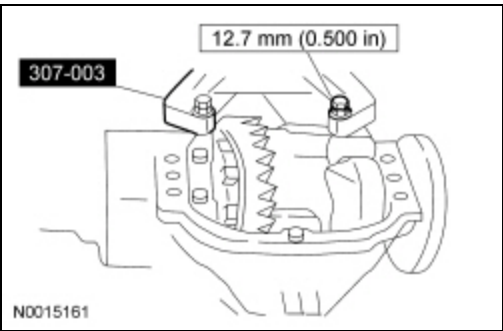
Removal

1. Remove the axle assembly. Refer to Axle Assembly in this section.
2. Remove the 10 bolts and the differential housing cover.
 - Drain the fluid.

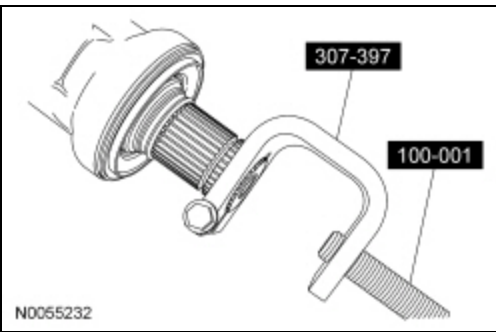


3. **NOTICE:** The fixture mounting bolts must have the minimum specified thread engagement in the carrier or damage to the component may occur.

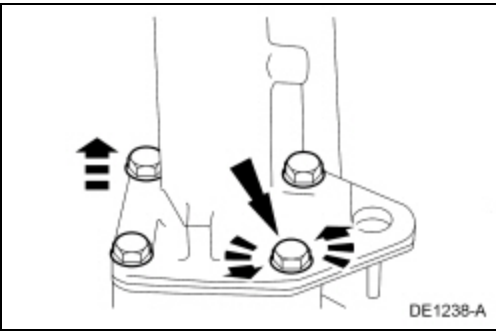
Install the Transmission Holding Fixture.



4. Using the Transmission Fluid Pump Remover and Slide Hammer, remove the intermediate shaft.

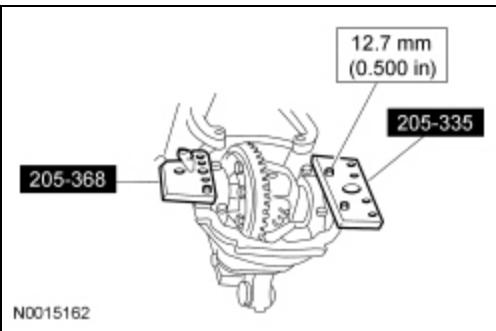


5. Remove the 4 bolts and the axle housing tube.



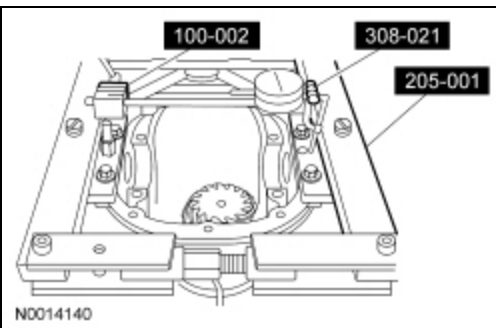
6. **NOTICE: The housing spreader adapter bolts must have the minimum specified thread engagement in the carrier or damage to the component may occur.**

Install the Adapter for Differential Housing Spreader and Differential Housing Spreader Plate.



7. Install the Dial Indicator Gauge with Holding Fixture, Clutch Housing Gauge and Differential Carrier Spreader.

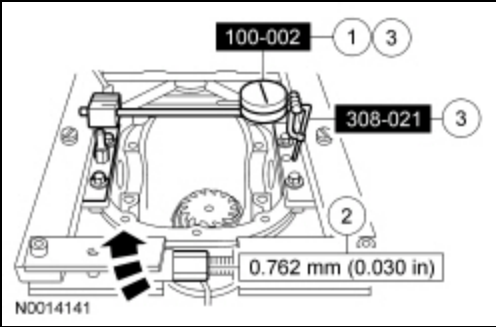
- Position the adapter tip in the spreader adapter hole.



8. **NOTICE: Overspreading may damage the axle housing.**

Spread the axle housing to the specification.

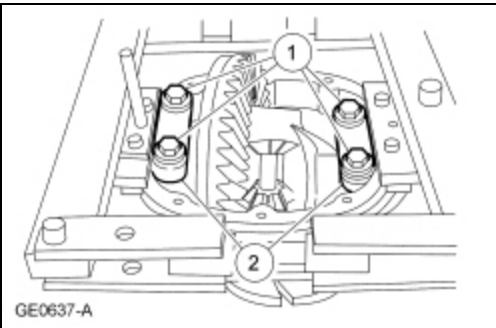
1. Adjust the Dial Indicator Gauge to zero.
2. Tighten the housing spreader screw to spread the axle housing to the specification.
3. Remove the Dial Indicator Gauge with Holding Fixture and Clutch Housing Gauge.



9. NOTICE: Mark the differential bearing caps Top or Bottom and LH or RH appropriately before removing them. Always install the bearing caps in their original location and position or damage to the component may occur.

Mark and remove the 2 differential bearing caps.

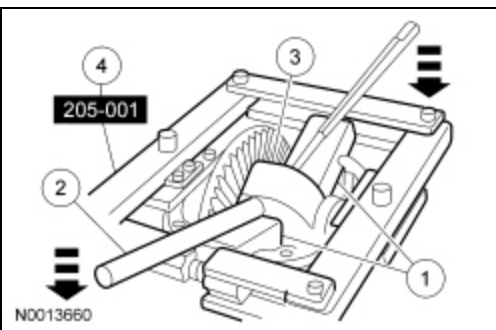
1. Remove the 4 bolts.
2. Remove the 2 differential bearing caps.



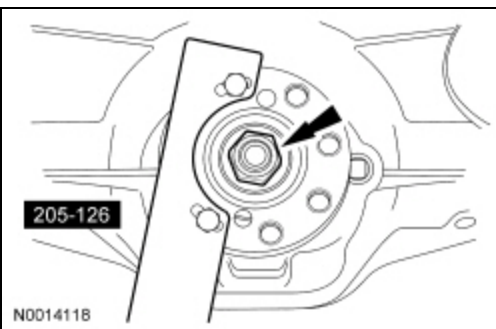
10. NOTICE: Use wood blocks to avoid axle housing damage.

Remove the differential assembly from the axle housing.

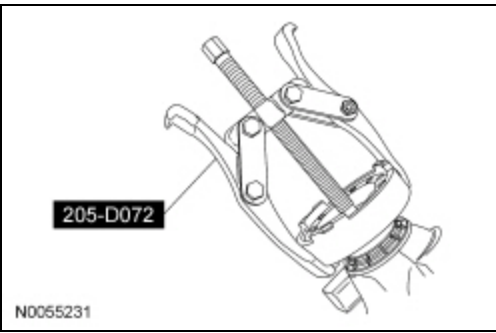
1. Position the wood blocks.
2. Position and push down on the 2 pry bars.
3. Remove the differential assembly.
4. Remove the Differential Carrier/Spreader.



11. Install the Drive Pinion Flange Holding Fixture and loosen, but do not remove, the pinion nut.

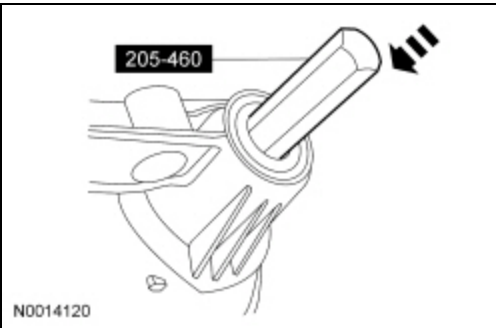


12. With the pinion nut still engaged by a few threads, use the 2 Jaw Puller to separate the axle U-joint flange from the pinion gear.

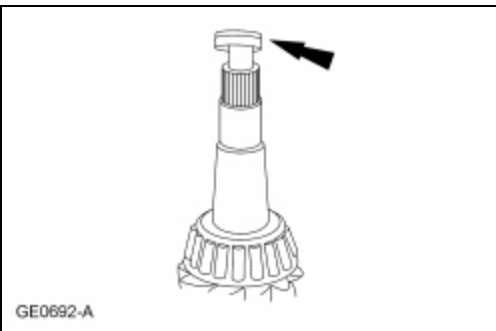


13. Remove the nut and the flange.

14. Using the Drive Pinion Thread Protector and a soft hammer, remove the pinion gear.



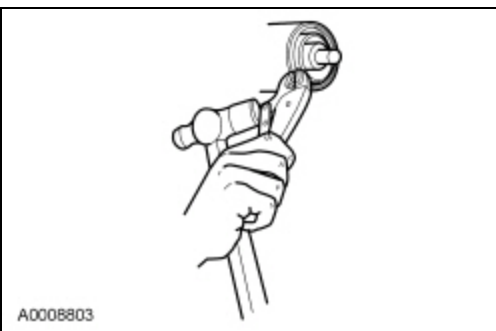
15. Discard the drive pinion collapsible spacer.



16. **NOTICE: Do not damage the carrier housing while removing the pinion seal.**

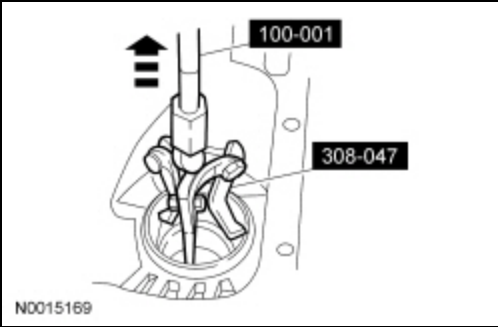
Remove the pinion seal and slinger.

- Pry up on the seal flange. Install gripping pliers and strike with a hammer to remove the seal.



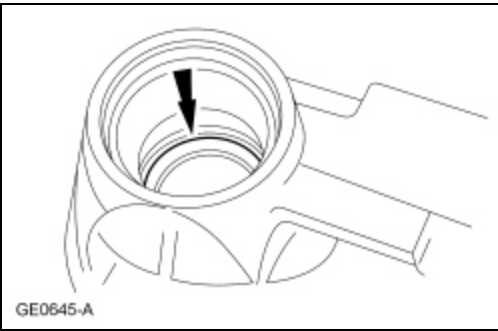
17. Remove the outer pinion bearing.

18. Using the Bearing Cup Remover and Slide Hammer, remove the inner pinion bearing cup.

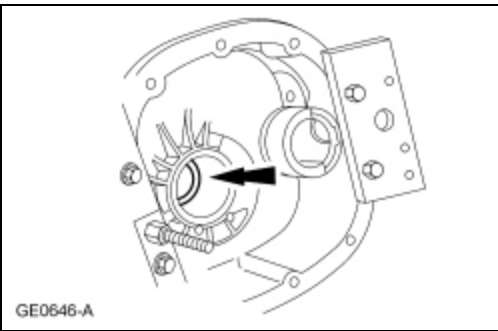


19. **NOTE:** The pinion bearing oil baffle aids in bearing lubrication. Normally, it is not necessary to remove the baffle.

If necessary, remove the pinion bearing oil baffle.

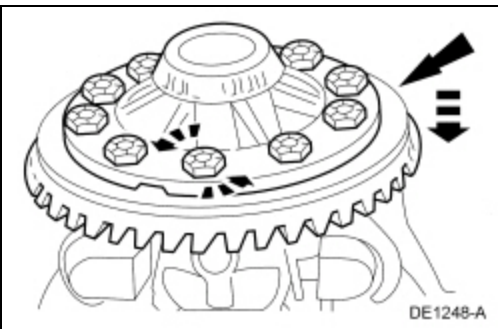


20. Remove the outer pinion bearing cup by tapping alternately on each side to prevent the cup from cocking.

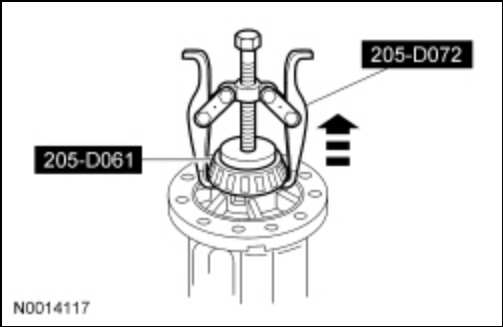


21. **NOTE:** Use a drift punch to separate the differential ring gear from the differential case.

Remove and discard the 10 differential ring gear bolts and remove the differential ring gear.



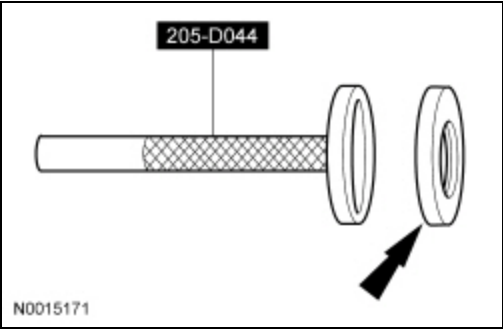
22. Using the 2 Jaw Puller and Step Plate, remove the 2 differential bearings.



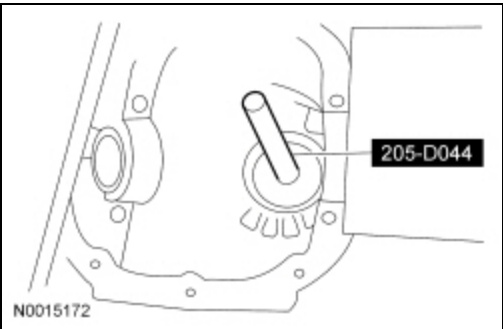
Installation

1. **NOTICE:** Do not omit the drive pinion bearing oil baffle or damage to the component may occur.

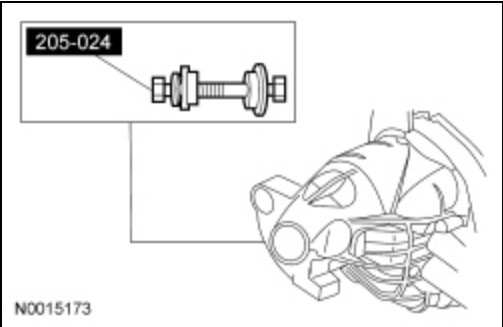
If removed, position the drive pinion bearing oil baffle on the Differential Carrier Bearings Installer.



2. Using the Differential Carrier Bearings Installer, install the drive pinion bearing oil baffle.



3. Using the Drive Pinion Bearing Cup Installer, install the drive pinion bearing cups.



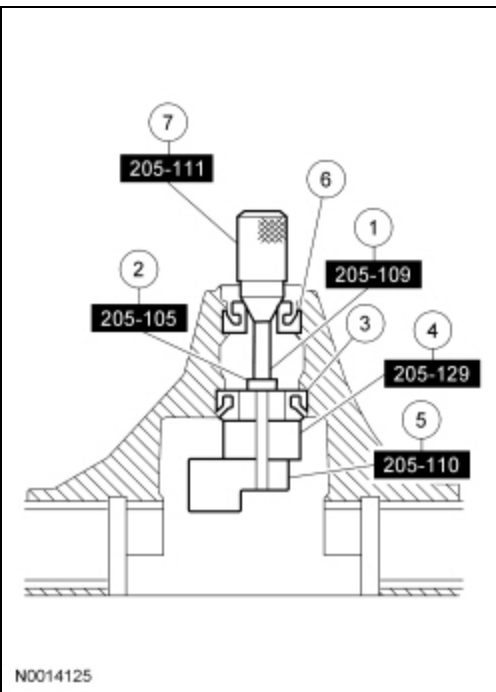
4. **NOTICE:** Use the same drive pinion bearings and the drive pinion bearing adjustment shim from the drive pinion bearing adjustment shim selection procedure for final assembly or damage to the component may occur.

NOTE: Install new drive pinion bearings without any additional lubricant since the anti-rust oil provides adequate lubricant without upsetting the drive pinion bearing preload settings.

Assemble and position the Adapters.

Item	Part Number	Description

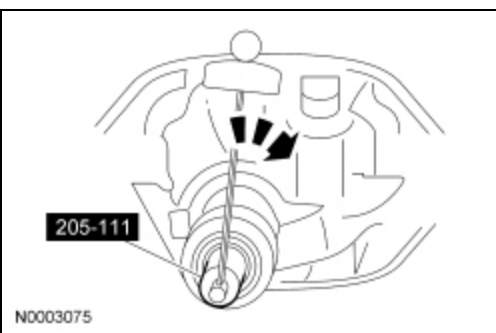
1	205-109-109	Adapter for 205-S127 (T76P-4020-A9)
2	205-105-105	Adapter for 205-S127 (T76P-4020-A3)
3	4630-4630	Drive pinion bearing (inner)
4	205-129-129	Adapter for 105-S127 (T79P-4020-A18)
5	205-110-110	Adapter for 205-S127 (T76P-4020-A10)
6	4621-4621	Drive pinion bearing (outer)
7	205-111-111	Adapter for 205-S127 (T76P-4020-A11)



5. **NOTE:** This step duplicates final drive pinion bearing preload.

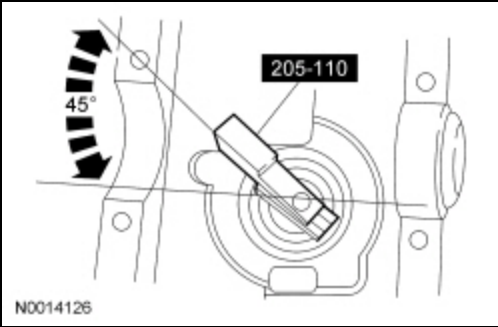
Tighten the Adapter.

- Tighten to 2.2 Nm (20 lb-in).



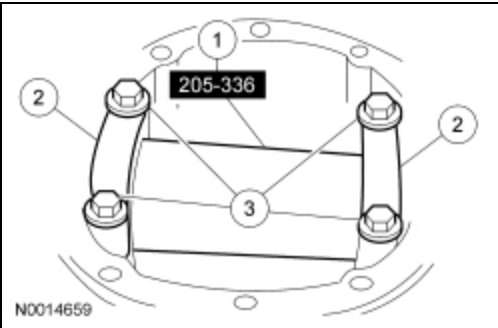
6. **NOTE:** The Adapter must be offset to obtain an accurate reading.

Rotate the Adapter several half-turns to make sure of correct seating of the drive pinion bearings and position the Adapter.



7. Install the Drive Pinion Gauge Tube.

1. Position the Drive Pinion Gauge Tube.
2. Install the 2 differential bearing caps.
3. Install the 4 differential bearing cap bolts.
 - Tighten to 105 Nm (77 lb-ft).

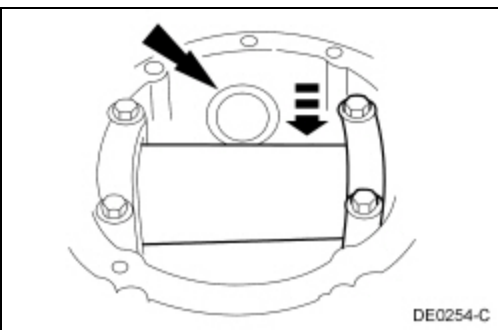


8. **NOTE:** Drive pinion bearing adjustment shims must be flat and clean.

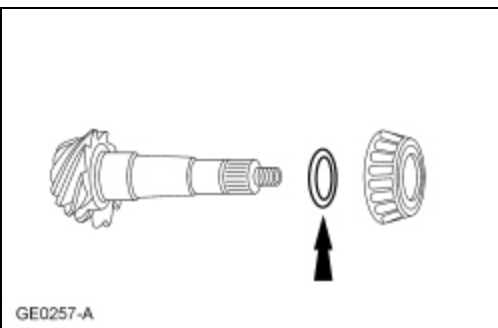
NOTE: A slight drag should be felt for correct drive pinion bearing adjustment shim selection. Do not attempt to force the drive pinion bearing adjustment shim between the gauge block and the gauge tube. This will minimize selection of a drive pinion bearing adjustment shim thicker than required, which results in a deep tooth contact in final assembly of integral axle assemblies.

Use a drive pinion bearing adjustment shim as a gauge for drive pinion bearing adjustment shim selection.

- After the correct drive pinion bearing adjustment shim thickness has been determined, remove all of the special tools.



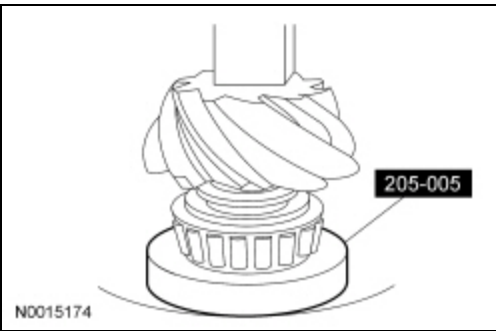
9. Position the drive pinion bearing adjustment shim and the inner pinion bearing on the new drive pinion gear stem.



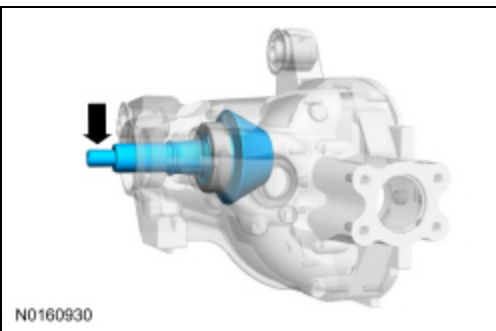
10. **NOTE:** *The same drive pinion bearings from the previous steps must be used.*

Install the inner drive pinion bearing.

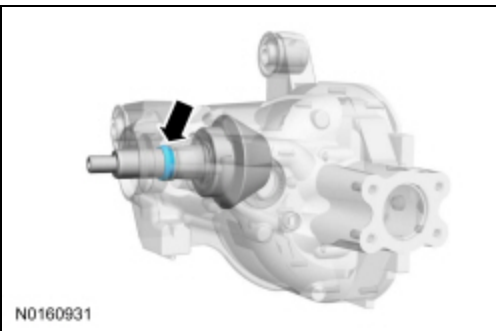
- Place the drive pinion bearing adjustment shim on the drive pinion gear.
- Using a suitable press and the Drive Pinion Bearing Cone Installer, press the inner drive pinion bearing until it is firmly seated on the pinion gear.



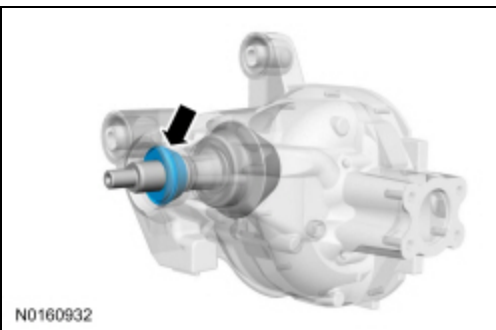
11. Install the pinion gear.



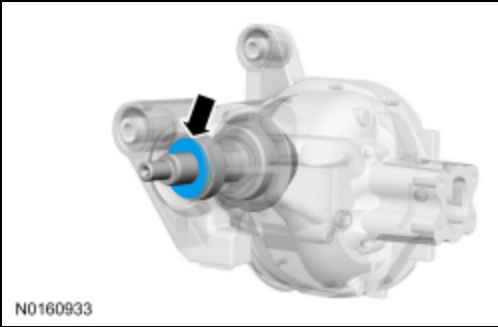
12. Install the new collapsible spacer.



13. Install the outer pinion bearing.



14. Install the pinion oil slinger.

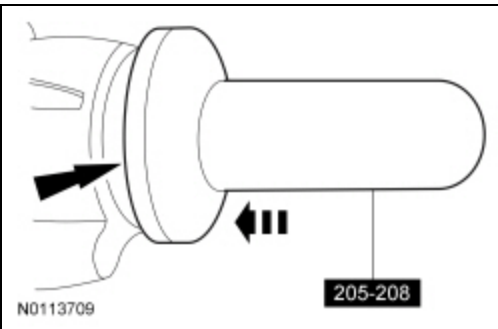


15. **NOTE:** *Coat the drive pinion seal lips with long-life grease.*

Install the drive pinion seal on the Drive Pinion Oil Seal Installer.

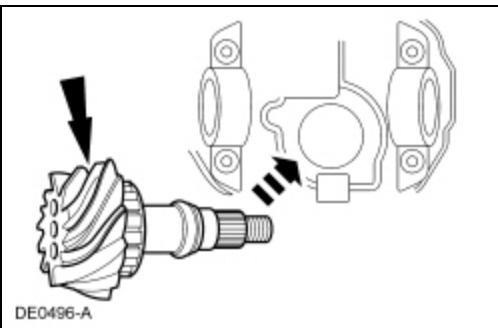
16. **NOTE:** *If the drive pinion seal becomes misaligned during installation, remove it and install a new drive pinion seal.*

Using the Drive Pinion Oil Seal Installer, install the drive pinion seal.

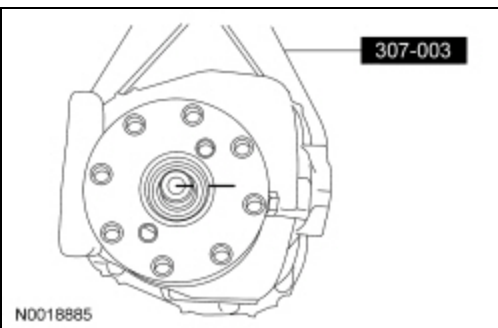


17. Lightly lubricate the drive pinion gear splines with axle lubricant.

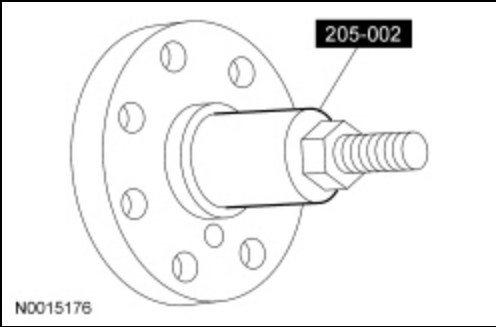
18. Position the drive pinion gear in the axle housing.



19. Position the pinion flange onto the pinion shaft while mounted on the Transmission Holding Fixture.



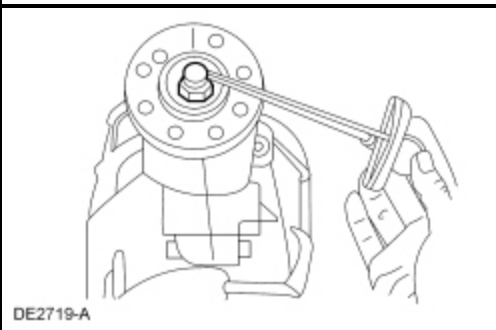
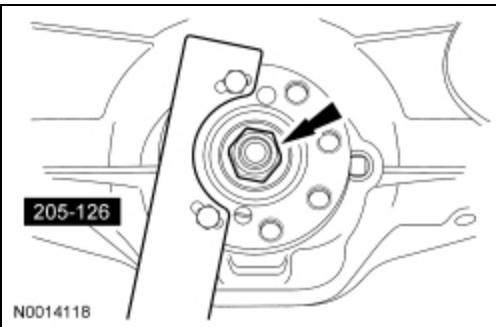
20. Using the Drive Pinion Flange Installer, install the pinion flange.



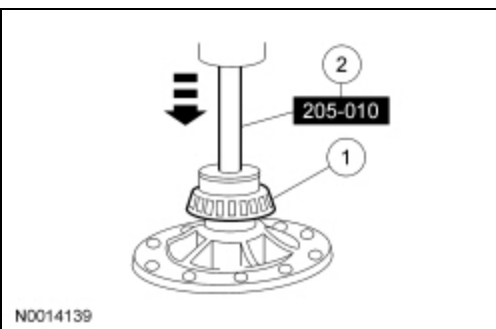
21. Apply a small amount of axle lubricant to the washer side of the new pinion nut and install the new pinion nut.
22. **NOTICE: Do not loosen the pinion nut to reduce drive pinion bearing preload under any circumstance. If it is necessary to reduce drive pinion bearing preload, install a new drive pinion collapsible spacer and pinion nut or damage to the component may occur.**

Install the Drive Pinion Flange Holding Fixture and tighten the pinion nut. For additional information, refer to the torque specifications in the Specifications portion of this section.

- Rotate the drive pinion gear occasionally to make sure the drive pinion bearings are seating correctly. Take frequent drive pinion bearing preload readings by rotating the drive pinion gear with a Nm (lb-in) torque wrench.

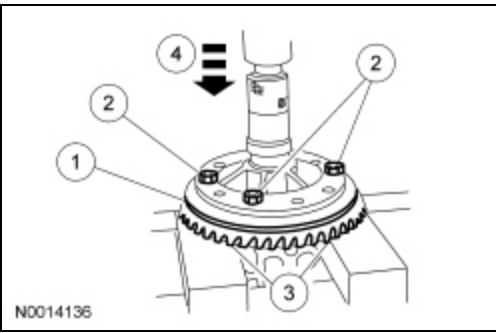


23. Press the left and right differential bearing on the differential carrier.
 1. Position the differential bearing.
 2. Using the Differential Side Bearing Installer, press the differential bearing on the differential carrier.
 - Repeat for the other side.

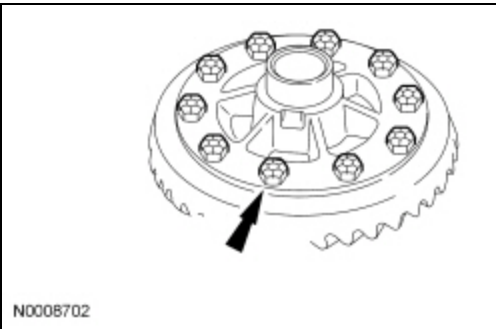


24. Install the differential ring gear.
 1. Place the differential ring gear on the differential assembly.
 2. Hand-start 3 new ring gear bolts to align the holes in the differential ring gear and the differential assembly.

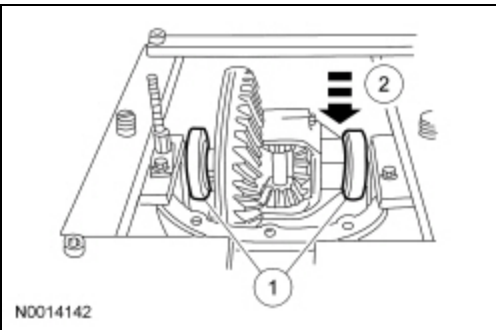
3. Place the differential carrier and differential ring gear onto the press bed blocks with the differential ring gear teeth facing downward.
4. Press the differential ring gear into place.



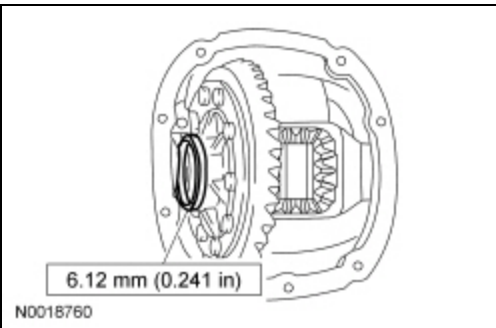
25. Install the 10 new ring gear bolts.
 - Tighten to 128 Nm (95 lb-ft).



26. Install the differential carrier.
 1. Position the differential bearing cups on the differential bearings.
 2. Lower the differential carrier in place.

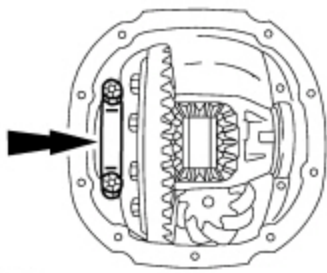


27. Install a differential bearing shim of the thickness shown on the LH side of the differential bearing.



28. **NOTE:** Apply pressure toward the left side to make sure the left bearing cup is seated.

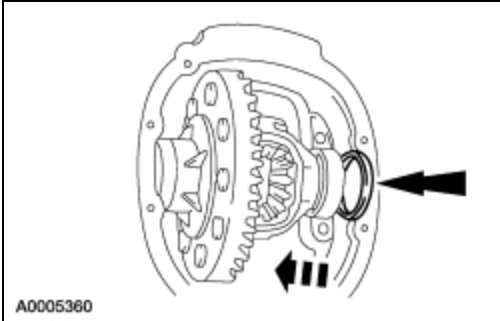
Install the left bearing cap and loosely install the 2 bearing cap bolts.



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29. **NOTE:** Apply pressure toward the left side to make sure the left bearing cap is seated.

Install progressively thicker differential bearing shims on the right side until the thickest differential bearing shim can be inserted by hand.



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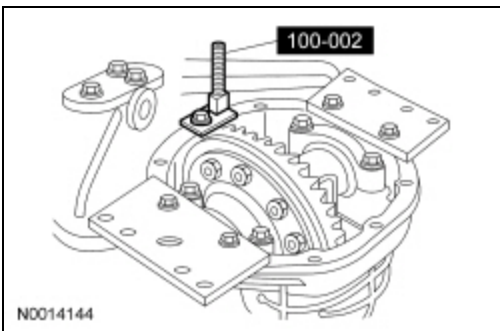
30. Install the right bearing cap and 2 bolts.

31. Tighten the 4 differential bearing cap bolts.

- Tighten to 105 Nm (77 lb-ft).

32. Rotate the differential carrier to make sure it turns freely.

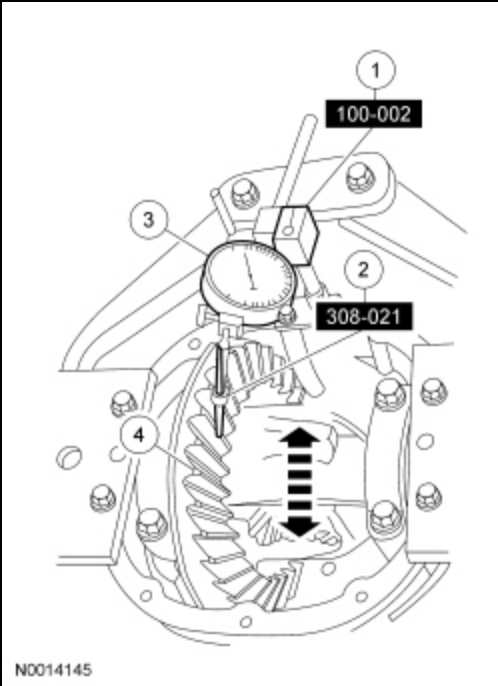
33. Install the Dial Indicator Gauge with Holding Fixture at the 12 o'clock position.



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34. Using the Dial Indicator Gauge with Holding Fixture and Clutch Housing Gauge, measure the differential ring gear backlash at 4 equally spaced points.

1. Attach the Dial Indicator Gauge with Holding Fixture and Clutch Housing Gauge.
2. Position the Dial Indicator Gauge with Holding Fixture and Clutch Housing Gauge tip centrally on a drive tooth.
3. Zero the Dial Indicator.
4. Turn the differential ring gear without turning the drive pinion gear. Record the indicator reading. The allowable backlash is 0.203 mm (0.008 in) to 0.305 mm (0.012 in) and must not vary more than 0.101 mm (0.004 in) between points measured. A backlash variation of more than 0.101 mm (0.004 in) between points checked indicates gear/case runout.
 - To correct for a high or low backlash, proceed as follows.



35. To correct for high or low backlash, increase the thickness of one differential bearing shim and decrease the thickness of the other differential bearing shim by the same amount. Refer to the following tables when adjusting the backlash.

Backlash Change Required		Thickness Change Required	
mm	Inch	mm	Inch
0.025	0.001	0.050	0.002
0.050	0.002	0.050	0.002
0.076	0.003	0.101	0.004
0.101	0.004	0.152	0.006
0.127	0.005	0.152	0.006
0.152	0.006	0.203	0.008
0.177	0.007	0.254	0.010
0.203	0.008	0.254	0.010
0.228	0.009	0.304	0.012
0.254	0.010	0.355	0.014
0.279	0.011	0.355	0.014
0.304	0.012	0.406	0.016
0.330	0.013	0.457	0.018
0.335	0.014	0.457	0.018
0.381	0.015	0.508	0.020

Numbers of Stripes and Color Code	Dimension A	
	mm	Inch
2 — C-COAL	7.7978- 7.8105	0.3070- 0.3075
1 — C-COAL	7.7470- 7.7597	0.3050- 0.3055
5 — BLU	7.6962- 7.7089	0.3030- 0.3035
4 — BLU	7.6454- 7.6581	0.3010- 0.3015
3 — BLU	7.5946- 7.6073	0.2990- 0.2995
2 — BLU	7.5458- 7.5565	0.2970- 0.2975
1 — BLU	7.493- 7.505	0.2950- 0.2955
5 — PINK	7.4422- 7.4549	0.2930- 0.2935
4 — PINK	7.3914- 7.4041	0.2910- 0.2915
3 — PINK	7.3406- 7.3533	0.2890- 0.2895
2 — PINK	7.2898- 7.3025	0.2870- 0.2875
1 — PINK	7.2390- 7.2517	0.2850- 0.2855
5 — GRN	7.1882- 7.2009	0.2830- 0.2835
4 — GRN	7.1374- 7.1501	0.2810- 0.2815
3 — GRN	7.0866- 7.0993	0.2790- 0.2795
2 — GRN	7.0358- 7.0485	0.2770- 0.2775
1 — GRN	6.9850- 7.0485	0.2750- 0.2755
5 — WH	6.9342- 6.9469	0.2730- 0.2735
4 — WH	6.8834- 6.8961	0.2710- 0.2715
3 — WH	6.8326- 6.8453	0.2690- 0.2695
2 — WH	6.7818- 6.7945	0.2670- 0.2675
1 — WH	6.7310- 6.7437	0.2650- 0.2655
5 — YEL	6.6802- 6.6929	0.2630- 0.2635
4 — YEL	6.6294- 6.6421	0.2610- 0.2615
3 — YEL	6.5786- 6.5913	0.2590- 0.2595
2 — YEL	6.5278- 6.5405	0.2570- 0.2575
1 — YEL	6.4770- 6.4897	0.2550- 0.2555
5 — ORNG	6.4262- 6.4389	0.2530- 0.2535
4 — ORNG	6.3754- 6.3881	0.2510- 0.2515
3 — ORNG	6.3246- 6.3373	0.2490- 0.2495
2 — ORNG	6.2738- 6.2865	0.2470- 0.2475
1 — ORNG	6.2223- 6.2357	0.2450- 0.2455
2 — RED	6.1722- 6.1849	0.2430- 0.2435
1 — RED	6.1214- 6.1341	0.2410- 0.2415

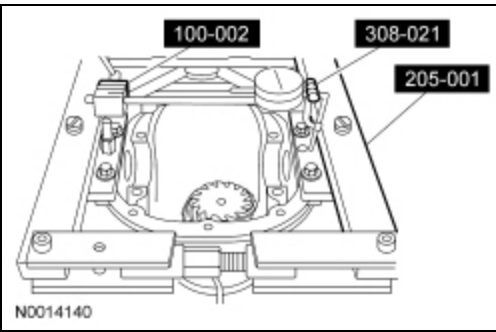
36. Remove the 4 differential bearing cap bolts.

- Remove the differential carrier.

37. Measure the thickness of the selected right and left bearing shims.

38. To establish differential bearing preload, increase both the left and right differential bearing shim thickness by 0.203 mm (0.008 in).

39. Install the Dial Indicator Gauge with Holding Fixture, Clutch Housing Gauge and Differential Carrier Spreader.

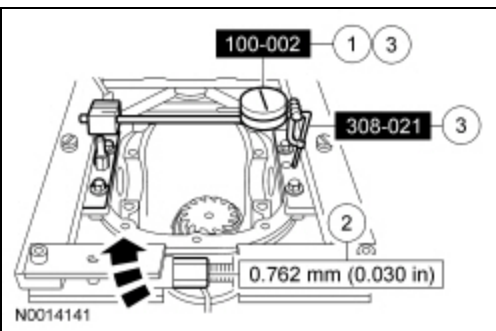


40. **NOTICE: Overspreading may damage the differential housing.**

NOTE: *Tighten and loosen the differential carrier spreader screw to normalize the housing spreader adapters prior to taking the final Dial Indicator reading.*

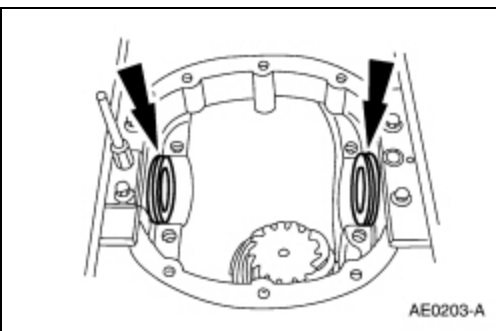
Spread the differential housing to the specification.

1. Adjust the Dial Indicator Gauge with Holding Fixture and Clutch Housing Gauge to zero.
2. Tighten the screw until spreading the differential housing to the specification.
3. Remove the Dial Indicator Gauge with Holding Fixture and Clutch Housing Gauge.



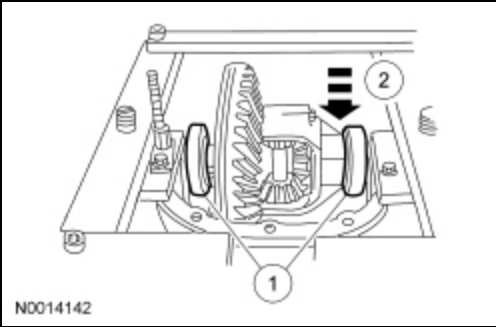
41. **NOTICE: To avoid galling the case, insert shims with a light coating of grease before installing the differential assembly. If not carried out in this order, the aluminum housing may be damaged.**

Place the differential bearing shims in the differential housing.



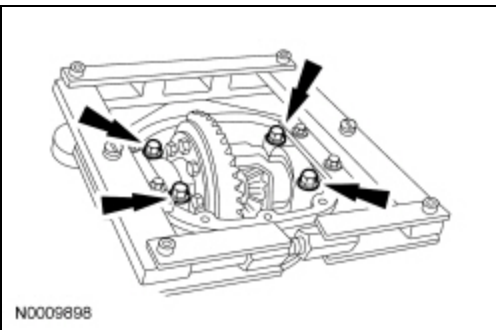
42. Install the differential carrier.

1. Position the differential bearing cups on the differential bearings.
2. Lower the differential carrier in place between the differential bearing shims.

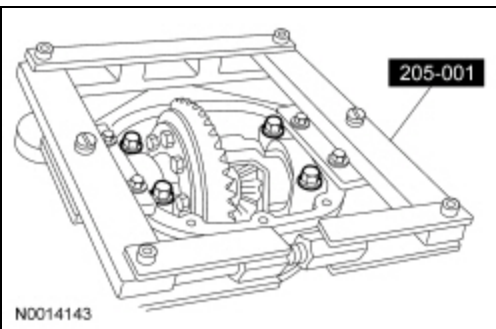


43. **NOTE:** *Hand-tighten the differential bearing cap bolts prior to releasing the carrier spreader.*

Install the 2 differential bearing caps and the 4 differential bearing cap bolts in their original locations and positions.



44. Loosen and remove the Differential Carrier Spreader.

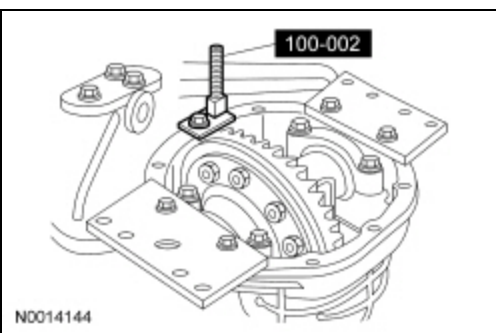


45. Tighten the 4 differential bearing cap bolts.

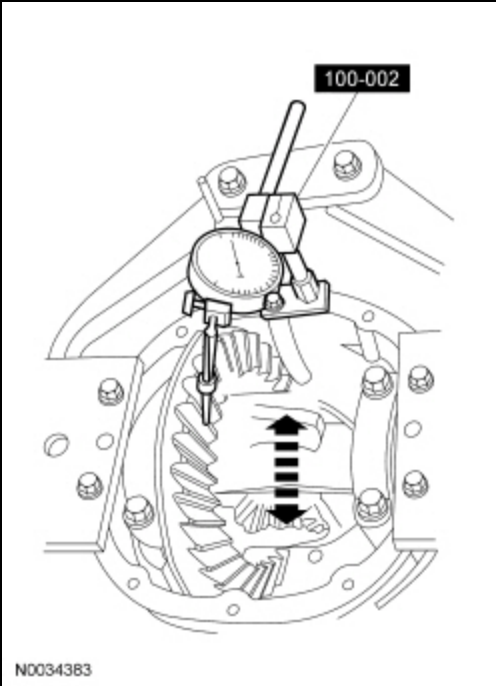
- Tighten to 105 Nm (77 lb-ft).

46. Rotate the differential carrier to make sure it turns freely.

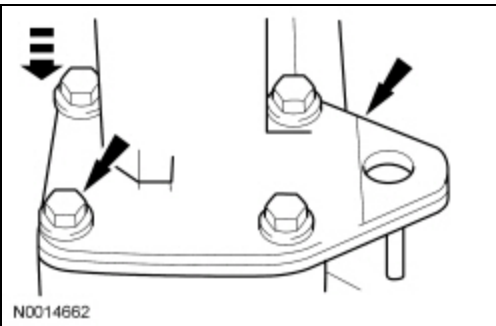
47. Install the Dial Indicator Gauge with Holding Fixture at the 12 o'clock position.



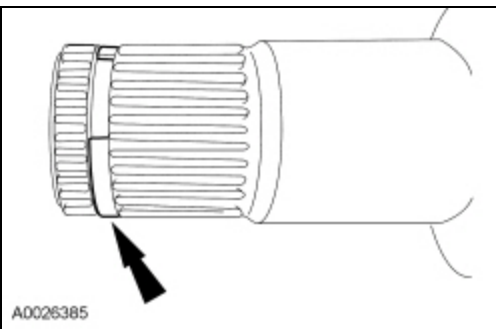
48. Using the Dial Indicator Gauge with Holding Fixture, recheck the ring gear backlash.



49. Apply marking compound and rotate the differential assembly 5 complete revolutions.
50. Verify an acceptable pattern check. For additional information, refer to Diagnosis and Testing in this section.
51. Install the 4 bolts and the axle housing tube.
 - Tighten to 73 Nm (54 lb-ft).



52. Install a new stub shaft seal. For additional information, refer to Stub Shaft Pilot Bearing and Seal in this section.
53. Install a new circlip on the intermediate shaft end.

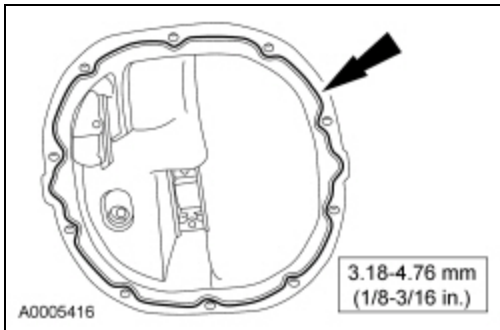


54. Insert the intermediate shaft and engage the circlip with the differential side gear.
55. Remove the axle assembly from the Transmission Holding Fixture and place it on the bench.
56. **NOTE:** Remove all of the silicone gasket and make sure the surfaces are free of oil before applying the new silicone gasket.

NOTE: The differential housing cover must be installed within 15 minutes of application of the silicone, or new sealant must be applied. If possible, allow one hour before filling with lubricant to make sure the silicone sealant has correctly cured.

Apply a continuous bead of sealant, of the specified thickness, to the differential housing cover mounting surface as shown in the illustration.

- Use silicone gasket and sealant.



57. Install the differential housing cover and 10 bolts.

- Tighten to 32 Nm (24 lb-ft).

58. Install the axle assembly. Refer to Axle Assembly in this section.

59. Remove the fill plug and fill the axle assembly with the specified amount and type of axle lubricant.

- Tighten the fill plug to 25 Nm (18 lb-ft).